

Load Balancing — Guided Notes ANSWER KEY

Unit 3.2.2 | CompTIA Network+ N10-009 Obj. 3.2 | N10-008 3.2

For instructor use / distribute after students complete the notes.

Question 1

A load balancer sits between _____ and a pool of _____. Clients connect to the load balancer's _____ (VIP) address and never communicate directly with individual backend servers.

Answer: Clients; backend servers; virtual IP (VIP) address.

Question 2

Layer 4 load balancing makes forwarding decisions based on _____ and _____ — it does not inspect application content. Layer 7 load balancing can inspect _____, URLs, and cookies, which allows routing decisions like 'send /api/ requests to one server pool and /static/ to another.'

Answer: Source/destination IP address and port numbers; HTTP headers.

Question 3

Match each scheduling algorithm to its description:

Round robin — _____

Least connections — _____

IP hash — _____

Weighted round robin — _____

Answer: Round robin — distribute requests sequentially across all servers; Least connections — send to the server with fewest active connections; IP hash — hash client IP to always route the same client to the same server; Weighted round robin — servers with higher weights receive proportionally more requests.

Question 4

A Layer 7 health check sends an actual _____ request (e.g., GET /health) and checks the _____ code. Why is this more reliable than a Layer 3/4 TCP health check?

Answer: HTTP request; HTTP status code (200 = healthy, 5xx = unhealthy). A TCP check only confirms the port is open — the process is running. A Layer 7 check confirms the application can actually respond correctly. A web server can accept TCP connections while returning 500 errors; only an HTTP health check catches that.

Question 5

Sticky sessions (session persistence) ensure requests from the same client always reach the _____ backend server. Why are sticky sessions a workaround rather than an ideal solution?

Answer: Same. Sticky sessions exist because the application stores session data locally on the server (in memory or disk). The ideal solution is a stateless application that stores session data in a shared backend (Redis, database) — then any server can handle any request and sticky sessions aren't needed. Sticky sessions reduce load balancing effectiveness and complicate failover.

Question 6

DNS round robin assigns multiple _____ records to the same hostname. List two significant limitations that make it unsuitable as the sole availability mechanism.

Answer: A records (or AAAA for IPv6). Limitations: (1) No health checking — DNS keeps returning a failed server's IP until the record is manually changed or the TTL expires; (2) DNS caching means clients stick to their resolved IP for the TTL duration, so distribution is uneven and failover is slow (minutes to hours, not seconds).

Question 7

Global Server Load Balancing (GSLB) distributes traffic across multiple _____ or _____. It typically works through _____ — returning different IP addresses based on client location or data center health.

Answer: Data centers; geographic regions. It works through DNS — the GSLB system acts as the authoritative DNS resolver and returns different IPs based on geography and health status.

Question 8 — Real World Application

REAL WORLD: A school's web portal is slow during exam week when all students log in simultaneously. The IT team wants to add load balancing. They have three identical servers available. Describe: which load balancing algorithm you would recommend, whether sticky sessions are needed, and what type of health check you would configure.

Answer: Algorithm: least connections — it handles variable request complexity (some students loading content, some submitting forms) better than round robin, which assumes uniform requests. Sticky sessions: depends on how the portal stores sessions. If the app stores session data in a shared database, skip sticky sessions. If it stores sessions locally, enable cookie-based sticky sessions. Health check: Layer 7 HTTP check against a /health endpoint that verifies the app and database are both responding — a TCP check isn't sufficient for an application with a database dependency.